Moira Bohley

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EDUCATION

University of California San Diego Exp. Graduation December 2026
M.S. Electrical Engineering (ISRC | Intelligent Systems Robotics & Controls)
University of California San Diego June 2025
B.S. Electrical Engineering, GPA: 3.375

PUBLICATIONS

[1] E. Huang, **M. Bohley**, et al., "Integrated Respiratory Sensing and Gated Control for Autonomous Needle Insertion via CT Bore Robotic Arm," *under review at IEEE ICRA 2027*. arXiv:2610.30452, **Co-first author**

[2] E. Peiros, **M. Bohley**, et al., "Bi-manual Stabilization of the Cervical Spine for Safe Physical Human-Robot Interaction in Rolling Maneuvers," *under review at IEEE ICRA 2027*. arXiv:2610.04220, **Second author**

SKILLS

Programming: Python, C/C++, MATLAB, Git, Docker
ML & Perception: PyTorch, Keras, OpenCV, Diffusion Models, Computer Vision, Fiducial Pose Tracking (ArUco)
Robotics: ROS 2, MuJoCo, Isaac Sim, Humanoid & Manipulator Control, Motion Planning (A*, RRT*, OMPL), SLAM, Hardware Integration, Microcontrollers
Controls & Estimation: Optimal Control, Model Predictive Control (MPC), Optimization-based IK, Force & Feedforward Control, Delay Compensation, EKF, Sensor Fusion

EXPERIENCE

UCSD Advance Robotics Controls (ARC) Lab, Graduate Researcher, June 2025 - Present
Search and Rescue Bimanual Cervical Spine Stabilization

- Created digital twin for bimanual robotic arm manipulation of 7-dof humanoid manipulators to maintain c-spine stability of a patient's head during movement using non-linear optimization Inverse Kinematics to solve for multiple objectives including position, orientation, and smoothness
- Devised head–torso pose tracking with ArUco markers and computer vision to measure spine biomechanics error
- Developed bimanual grasp controller by estimating force from joint-torque feedback to regulate grip
- Integrated system onto G1 humanoid platform for real-world experiments; second-author ICRA 2027 submission [2]

CT Surgical Arm

- Led co-first-author ICRA 2027 submission [1], coordinating a 7-person team across mechanical and software efforts
- Developed algorithms to locate optimal needle insertion points and orientations to avoid bone collision and excessive tissue damage before reaching goal
- Implemented simulation and physical testing of robot arm through ROS 2
- Used tactile sensing in an EKF to estimate chest motion for accurate needle insertion
- Developed control scheme in order to compensate for delay in input signals and motor actuation to precisely insert needle at proper times for patient safety in the breathing cycle

PROJECTS

IMU-Lidar Based SLAM

- Fused IMU and LiDAR data to estimate trajectory via the Kabsch algorithm and build occupancy grid maps

Path Planning Algorithms

- Implemented A*, RRT, and RRT* from scratch and with OMPL (Python/C++) for obstacle-rich environments
- Transferred RRT* algorithms to simulation control of Franka Panda arm in order to train Motion Planning Neural Networks for solving manipulator paths in random obstacle rich environments

Database Implementation

- Built a relational database in C++ from scratch that interprets SQL commands, using command patterns and system design

Diffusion models on Medical Images

- Modified and trained the SRDiff super-resolution diffusion model on medical images to deblur 2D and 3D CT scans

Motion Control Planning

- Designed MPC and infinite-horizon optimal control policies for differential-drive trajectory tracking and obstacle avoidance